

PVsyst - Simulation report

Grid-Connected System

Project: New Project

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 5.50 kWp

São Carlos - Brazil



Project: New Project

Variant: New simulation variant

PVsyst V8.1.4

VCO, Simulation date:
17/07/26 10:55
with V8.1.4

Project summary

Geographical Site

São Carlos
Brazil

Situation

Latitude -22.0180 °(S)
Longitude -47.8912 °(W)
Altitude 854 m
Time zone UTC-3

Project settings

Albedo 0.20

Weather data

São Carlos
NASA POWER satellite data 2001-2020 - Synthetic

System summary

Grid-Connected System

Orientation #1

Fixed plane

Tilt/Azimuth 28 / 0 °

System information

PV Array

Nb. of modules 10 units
Pnom total 5.50 kWp

No 3D scene defined, no shadings

Near Shadings

no Shadings

User's needs

Unlimited load (grid)

Inverters

Nb. of units 1 unit
Total power 5 kWac
Pnom ratio 1.10

Results summary

Simulation step Hourly

Produced Energy 9322.7 kWh/year Specific production 1695 kWh/kWp/year Perf. Ratio PR 84.04 %

Table of contents

Project and results summary	2
General parameters, PV Array Characteristics, System losses	3
Main results	4
Loss diagram	5
Predef. graphs	6
Single-line diagram	7



Project: New Project

Variant: New simulation variant

PVsyst V8.1.4

VC0, Simulation date:
17/07/26 10:55
with V8.1.4

General parameters

Grid-Connected System

No 3D scene defined, no shadings

Simulation step

Hourly

Orientation #1

Fixed plane

Tilt/Azimuth 28 / 0 °

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

no Shadings

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer Generic
Model LR5-72HPH-550M G2
(Original PVsyst database)
Unit Nom. Power 550 Wp
Number of PV modules 10 units
Nominal (STC) 5.50 kWp
Modules 2 units x 5 In series

At operating cond. (50°C)

Pmpp 5.04 kWp
U mpp 189 V
I mpp 27 A

Total PV power

Nominal (STC) 6 kWp
Total 10 modules
Module area 25.8 m²
Cell area 23.8 m²

Inverter

Manufacturer Generic
Model SUN2000-5KTL-L1
(Original PVsyst database)
Unit Nom. Power 5.00 kWac
Number of inverters 2 * MPPT 50% 1 unit
Total power 5.0 kWac
Operating voltage 80-600 V
Max. power (=>40°C) 5.50 kWac
Pnom ratio (DC:AC) 1.10
No power sharing between MPPTs

Total inverter power

Total power 5 kWac
Number of inverters 1 unit
Pnom ratio 1.10

Array losses

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 117 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss

Loss Fraction -0.75 %

Module mismatch losses

Loss Fraction 2.00 % at MPP

IAM loss factor

Incidence effect (IAM): User defined profile

0°	25°	45°	60°	65°	70°	75°	80°	90°
1.000	1.000	0.995	0.962	0.936	0.903	0.851	0.754	0.000



PVsyst V8.1.4

VC0, Simulation date:
17/07/26 10:55
with V8.1.4

Main results

Simulation step

Hourly

System Production

Produced Energy

9322.7 kWh/year

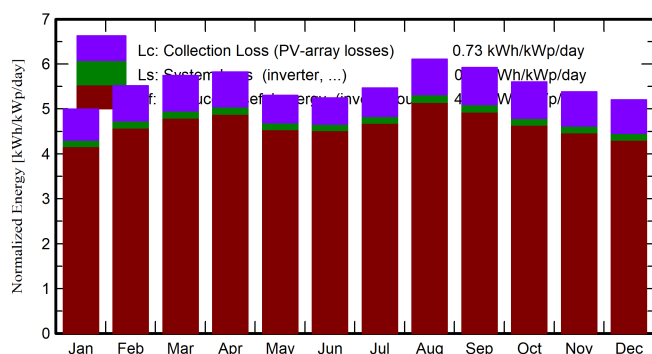
Specific production

1695 kWh/kWp/year

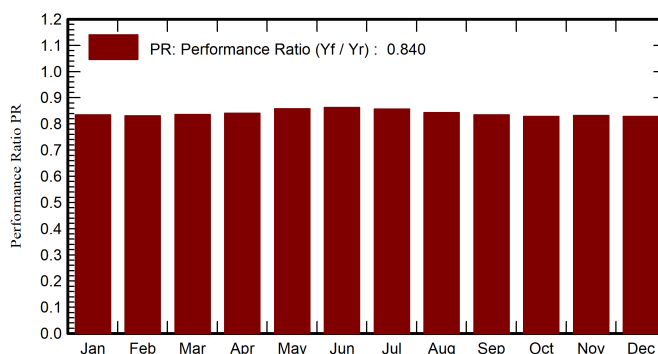
Perf. Ratio PR

84.04 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m ²	kWh/m ²	°C	kWh/m ²	kWh/m ²	kWh	kWh	ratio
January	175.8	83.60	23.80	154.8	151.1	734.8	710.5	0.835
February	163.6	66.90	23.81	154.5	151.6	730.2	706.3	0.831
March	169.7	64.90	23.20	178.1	174.8	845.8	818.5	0.835
April	147.9	47.80	21.81	174.6	172.0	833.7	806.9	0.840
May	125.2	41.70	18.89	164.3	162.2	800.4	774.8	0.858
June	112.9	33.90	18.24	157.4	155.4	770.9	746.6	0.862
July	124.7	34.10	18.31	169.6	167.5	825.1	799.1	0.857
August	150.0	36.90	20.28	189.4	186.9	907.2	878.6	0.843
September	159.3	49.10	22.86	177.6	174.6	841.9	815.2	0.835
October	178.9	69.90	24.33	173.8	169.8	817.9	791.8	0.829
November	180.4	78.40	23.89	161.5	157.4	763.7	738.7	0.832
December	188.3	85.89	24.09	161.4	156.8	760.8	735.6	0.829
Year	1876.7	693.09	21.95	2016.9	1980.2	9632.4	9322.7	0.840

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

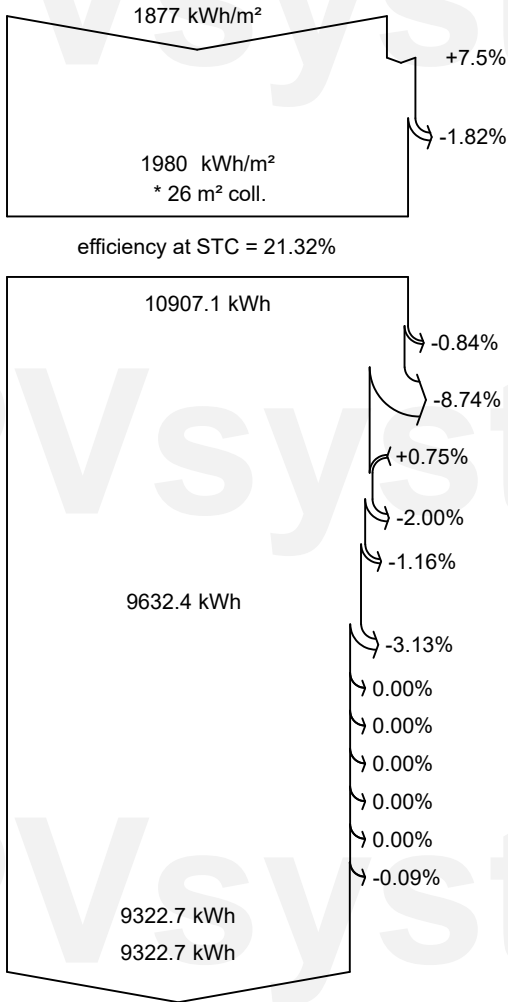
PR Performance Ratio



PVsyst V8.1.4

VC0, Simulation date:
17/07/26 10:55
with V8.1.4

Loss diagram



Global horizontal irradiation
Global incident in coll. plane

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

PV loss due to irradiance level

PV loss due to temperature

Module quality loss

Module array mismatch loss

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

Energy injected into grid

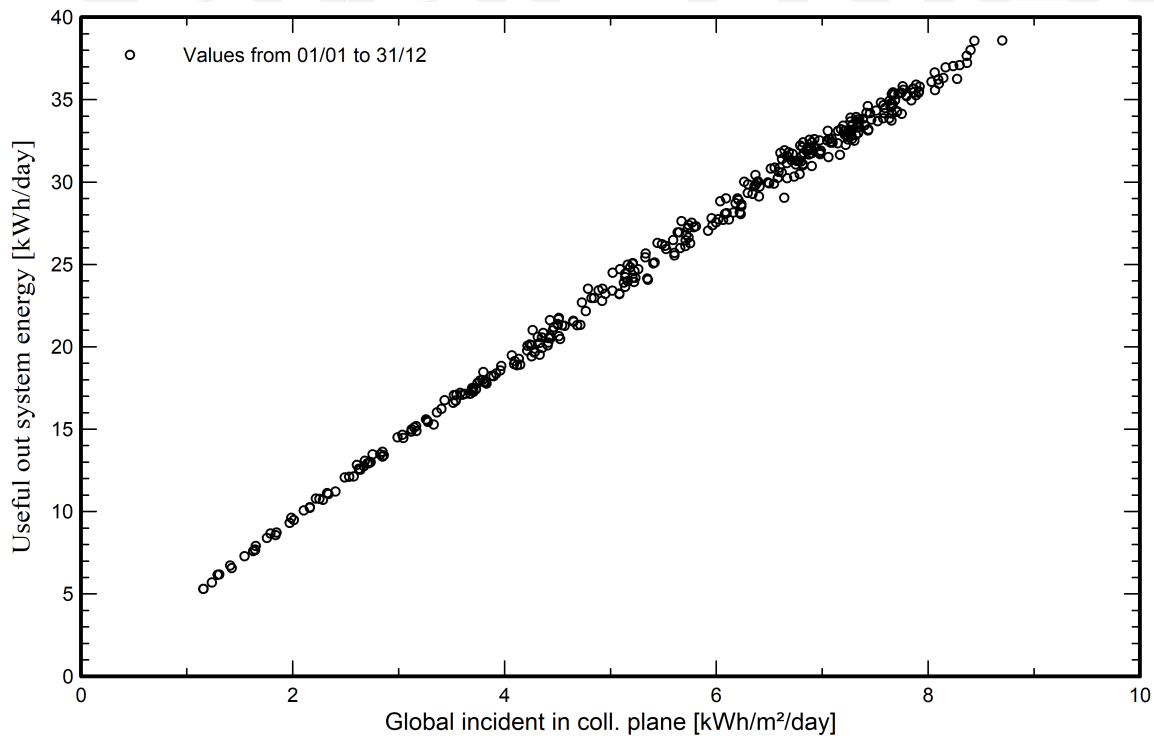


PVsyst V8.1.4

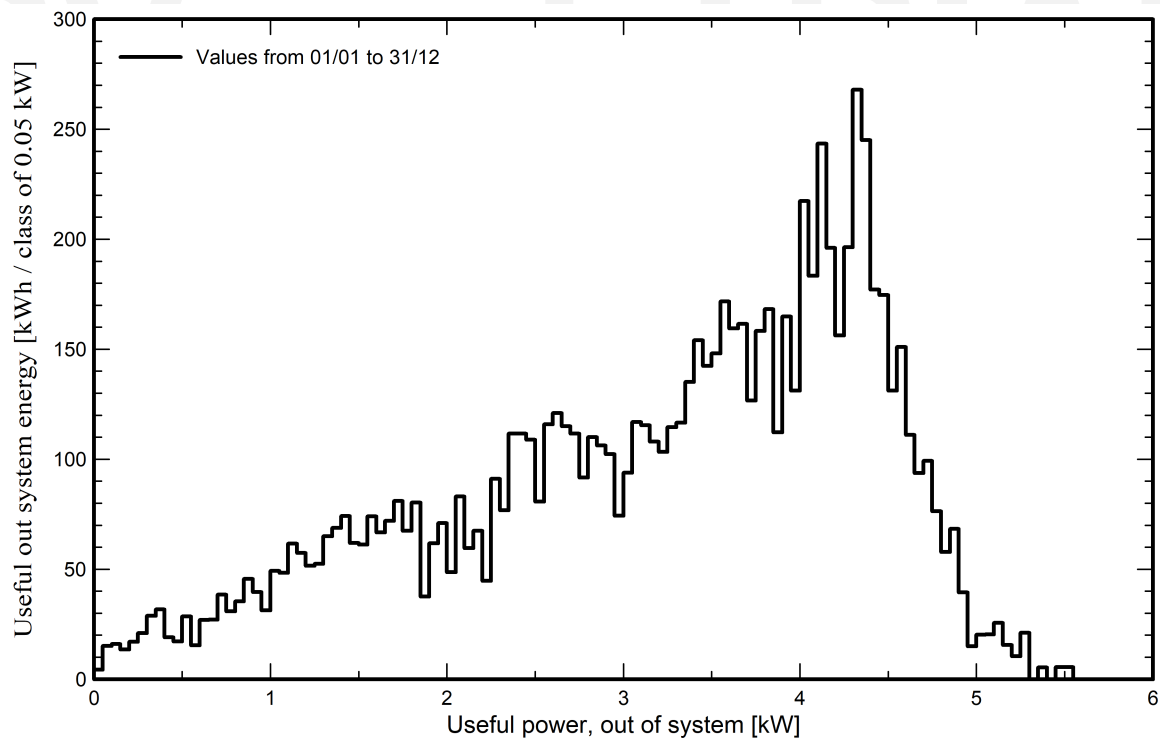
VC0, Simulation date:
17/07/26 10:55
with V8.1.4

Predef. graphs

Daily Input/Output diagram



System Output Power Distribution

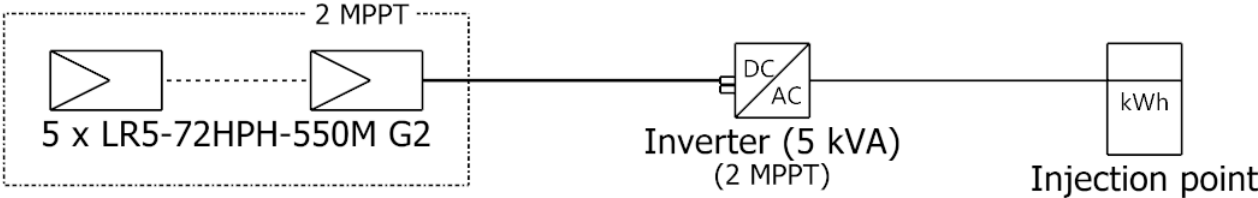




PVsyst V8.1.4

VC0, Simulation date:
17/07/26 10:55
with V8.1.4

Single-line diagram



PV module	LR5-72HPH-550M G2
Inverter	SUN2000-5KTL-L1
String	5 x LR5-72HPH-550M G2

New Project

VC0 : New simulation variant

17/07/26